

Lease first; preserve a 10 MW option

Investment committee memo | Initial baseline | 5 October 2026

Decision requested: Return the full 25 MW build for more evidence. Approve a capped £2m feasibility mandate within the facility allowance and competitive leasing procurement. A 10 MW owned module remains conditional; do not authorise construction or a GPU purchase order yet.

1. Need and scale

The brief states 25 MW is total site demand and that no institution has signed a long-term computing commitment [S1]. Our illustrative UK Midlands search region is not a secured site. A 20 MW IT load at PUE 1.25 gives 25 MW and 219 GWh/year at continuous full load. These are an envelope, not measured annual consumption.

The base hypothesis is 91.104m productive GPU-hours/year: 55m for training, 20m for teaching and 16.104m for inference. It assumes 16,000 benchmark-equivalent GPUs at 65% productive utilisation, with 1.25 kW of shared IT power per GPU. Failed jobs and idle reservations do not count as productive. Obtain workload benchmarks and signed commitments before sizing the build.

2. Comparable ten-year economics

All options deliver the same useful demand over service years 1-10. Owned capacity opens in year 3; leases bridge the first two years. Facility/grid capex is separate from the GPU fleet. GPU replacement occurs every four operating years. Values are constant GBP; costs and tariffs are illustrative assumptions.

Base case	Build 25 MW	Lease	Hybrid 10 MW
Cash before opening	£1158.7m	£0.0m	£737.4m
First operating-year opex	£50.1m	£228.3m	£144.5m
10-year present cost	£1552.9m	£1531.6m	£1478.7m
£ / productive GPU-hour	2.54	2.51	2.42
Gross capital at risk	£700.2m	£0.0m	£280.1m

Present cost uses an 8% discount rate and includes facility borrowing interest at 6%. Debt funds 60% of facility/grid capex; GPUs are cash-funded. Equity cash includes drawdowns and principal repayment; project costs do not count principal twice. Residual debt is repaid in year 10. No terminal salvage, tax, grant or inflation benefit is assumed.

Leasing is £2.50 per productive equivalent GPU-hour, assumed monthly cancellable. Construction is £10m per site MW; grid upgrades £30m for 25 MW; GPUs £25,000 each including host share. Hybrid serves up to 75% local utilisation, with demand above that leased. No claim is made that hybrid is always the least-cost option.

Fund evidence before committing capital

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3. Downside cases and resilience

Build 25 MW stress	Cash pre-open	Annual opex	£ / GPU-h	Capital risk
Base	£1158.7m	£50.1m	2.54	£700.2m
Grid +1 year	£1398.0m	£50.1m	2.71	£710.2m
Half utilisation	£930.9m	£43.2m	4.30	£700.2m

The delay case opens in year 4, defers GPUs and adds bridge leasing and carry costs. The half-utilisation case halves useful demand while owned capacity remains installed. Capital at risk is gross irreversible facility, grid, fleet and pre-opening interest, assuming zero recovery; it is not expected loss. Opex is the first owned operating year; full comparisons and year-by-year cash are on the site.

Reference resilience: two 30 MVA transformer paths, each rated for full load after derating; UPS usable energy of 4.17 MWh bridges ten minutes; six 5 MW generator modules provide N+1 at 25 MW. At 0.25 L/kWh plus 20% reserve, 48 hours needs 360,000 L fuel. Cooling remains on backup. No annual renewable certificate or intermittent generation receives firm-capacity credit. Engineer verification, permits and integrated outage tests remain outstanding.

4. Ownership, finance and fair access

The consortium SPV would own facility infrastructure and the committed GPU base; contract design/build, support, utility supply, fibre and burst compute. Development equity requires a capped budget and exit rights; construction debt needs binding grid terms, permits, fixed-scope bids and take-or-pay member contracts. Equipment finance is an alternative requiring a separate quote, not part of the cash-funded GPU base model.

Residual grid-delay and obsolescence risk stays with the SPV; utility/EPC remedies count only if contracted. Anchor-member withdrawal requires notice, credit support and exit payments. Proposed allocation: 60% reservations, 20% peer-reviewed research, 20% teaching/small-member access; cap shared-pool use by one member at 25% without board approval. Publish two-part tariffs, metering and independent cost reconciliation.

5. Three findings most likely to reverse the recommendation

Demand: Binding 3+ year reservations covering at least 70% of the 10 MW module's 42m productive GPU-hours, backed by benchmarks and credit support.

Grid: A deliverable connection offer with energisation milestones, total tariff, upgrades and N–1 engineering. A costly delay could favour permanent leasing.

Supplier economics: Performance-matched lease bids and independent construction/GPU quotes. A lower effective lease price may remove the case for ownership.

Sources: [S1] Course brief (Google Doc linked on site). [S2] DESNZ, DUKES 2025, 2024 generation: 50.4% renewable, historical edition. [S3] University of Bristol, Isambard-AI launch, 17 July 2025. Existing AIRR capacity is an alternative to investigate, not guaranteed access. Full source URLs and dates appear in the evidence library.